

MR_IL6R_CRC v46 - GEO validation integration and revised JTM package

Output Beijing time: 2026-05-17 03:18

Executive conclusion

The v45 GEO validation outputs provide usable external validation data for IL6R expression extraction and external survival consistency checks. The external validation does not support a robust prognostic association between IL6R expression and OS/DFS/DSS in GSE17536 or GSE17537, consistent with repaired TCGA OS/PFI analysis. Tumor-normal validation in GEO was not claimed because no cohort had an automatically reliable tumor/normal split; GSE39582 remained unavailable because of download timeout/parse failure.

Key extracted GEO numbers

Cohort	N	Primary probe	Median	IQR	Status
GSE39582	0	not_available			parse_failed/download_timeout
GSE17536	177	205945_at	6.649	6.292-7.012	IL6R extracted
GSE17537	55	226333_at	6.687	6.129-7.067	IL6R extracted
GSE14333	290	217489_s_at	2.908	1.969-4.061	IL6R extracted

External survival validation

Cohort	Endpoint	N	Events	HR	95% CI	P
GSE17536	OS	177	73	0.897	0.708-1.136	0.366
GSE17536	DFS	145	36	1.003	0.722-1.392	0.987
GSE17536	DSS	177	55	0.922	0.703-1.210	0.560
GSE17537	OS	55	20	1.327	0.864-2.037	0.196
GSE17537	DFS	42	6	0.563	0.204-1.554	0.267

Remaining limitations

- Repository URL/DOI remains pending and should be added before final submission.
- GSE39582 remained unavailable in the current automated run.

- GEO tumor-normal validation was not claimed because metadata were not reliably separable by automated rules.
- External survival analyses are univariable validation checks, not fully adjusted clinical prognostic models.